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critical constants are related to Van der Waals constants as follows

$$T_C = \frac{8\,a}{27\,R\,b}, \quad P_C = \frac{a}{27b^2} \text{ and } V_C = 3b$$

When a gas is made to expand adiabatically from a region of high pressure into a region of low pressure, the temperature of the gas is reduced rapidly and this is known as Joule-Thomson effect. This effect is used in the liquefication of gases.